

ORGANISASI & ARSITEKTUR KOMPUTER

Pararel Processing

Multiple Processor Organization

- Single instruction, single data stream - SISD
- Single instruction, multiple data stream - SIMD
- Multiple instruction, single data stream - MISD
- Multiple instruction, multiple data stream- MIMD

Single Instruction, Single Data Stream - SISD

- Single processor
- Single instruction stream
- Data stored in single memory
- Uni-processor

Single Instruction, Multiple Data Stream - SIMD

- Single machine instruction
- Controls simultaneous execution
- Number of processing elements
- Lockstep basis
- Each processing element has associated data memory
- Each instruction executed on different set of data by different processors
- Vector and array processors

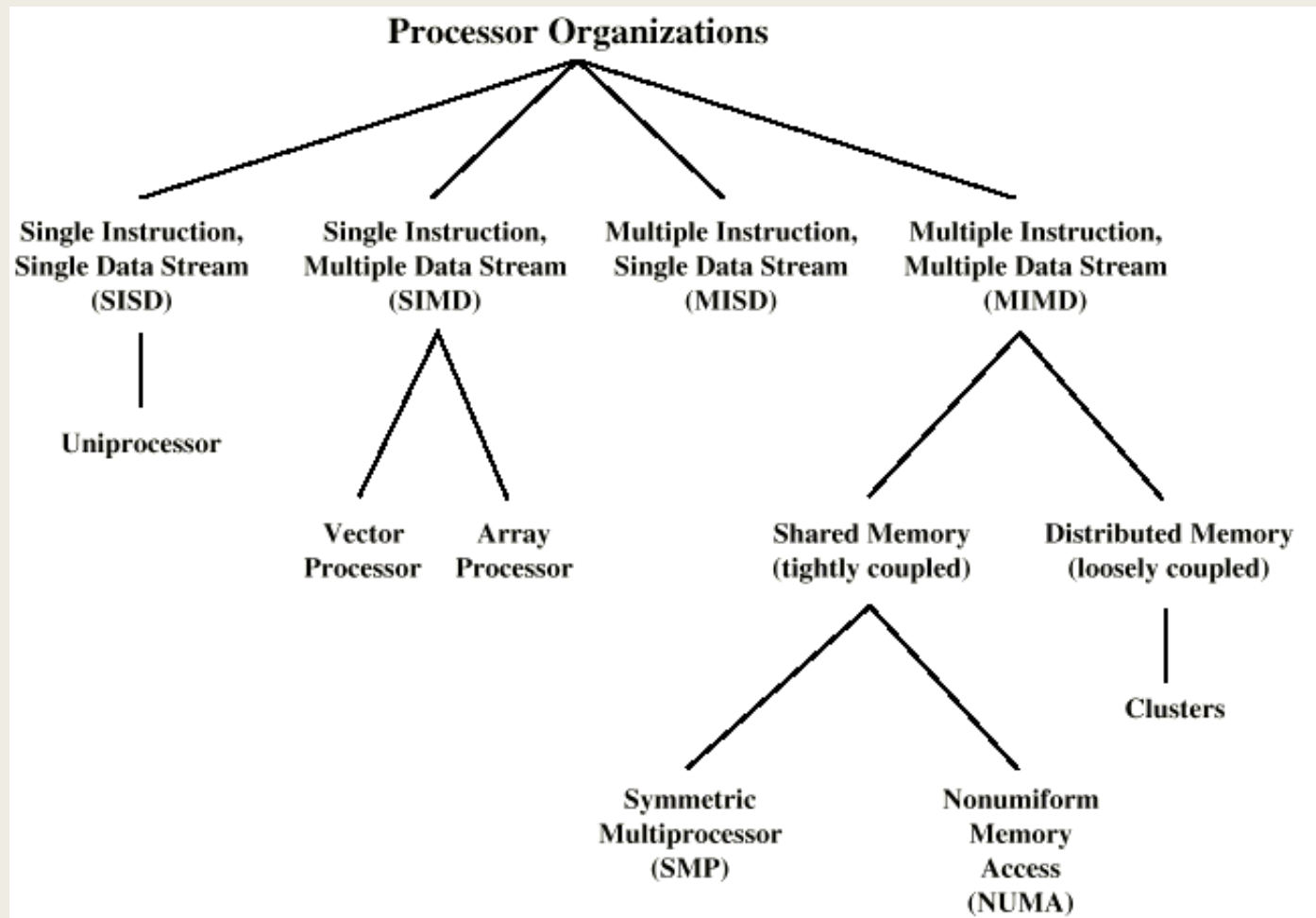
Multiple Instruction, Single Data Stream - MISD

- Sequence of data
- Transmitted to set of processors
- Each processor executes different instruction sequence
- Never been implemented

Multiple Instruction, Multiple Data Stream- MIMD

- Set of processors
- Simultaneously execute different instruction sequences
- Different sets of data
- SMPs, clusters and NUMA systems

Taxonomy of Parallel Processor Architectures



MIMD - Overview

- General purpose processors
- Each can process all instructions necessary
- Further classified by method of processor communication

Tightly Coupled - SMP

- Processors share memory
- Communicate via that shared memory
- Symmetric Multiprocessor (SMP)
 - *Share single memory or pool*
 - *Shared bus to access memory*
 - *Memory access time to given area of memory is approximately the same for each processor*

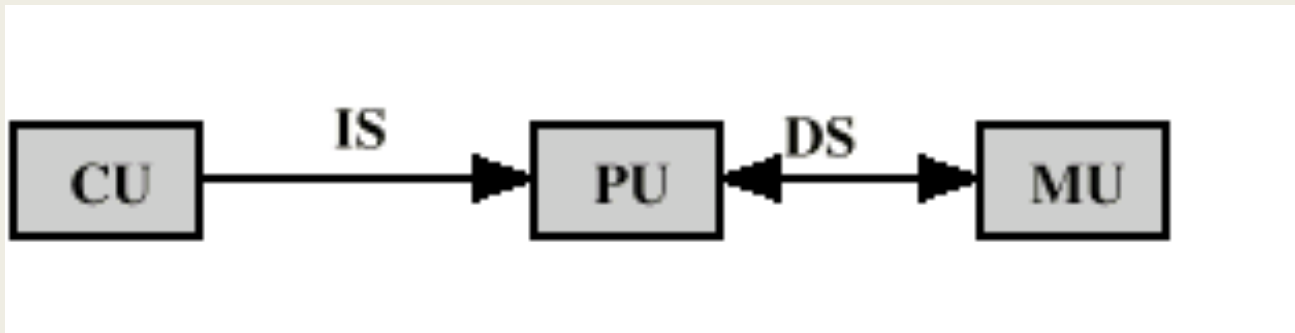
Tightly Coupled - NUMA

- Nonuniform memory access
- Access times to different regions of memory may differ

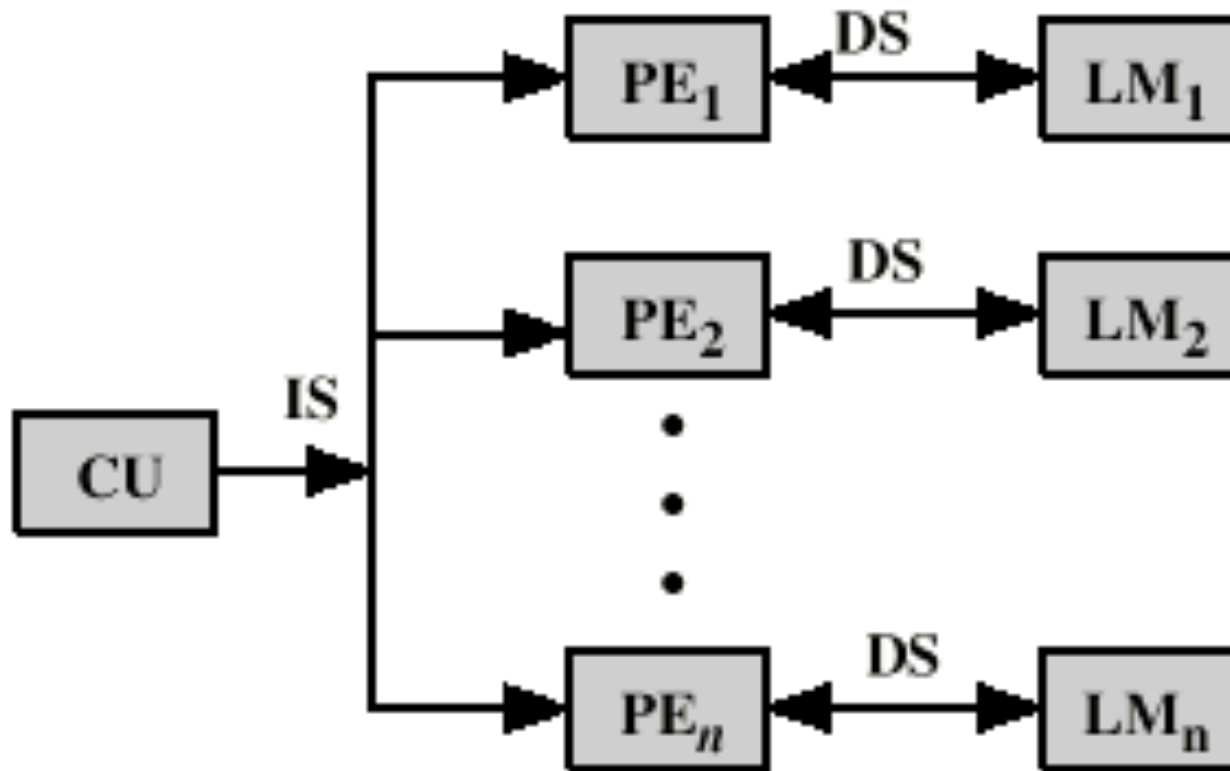
Loosely Coupled - Clusters

- Collection of independent uniprocessors or SMPs
- Interconnected to form a cluster
- Communication via fixed path or network connections

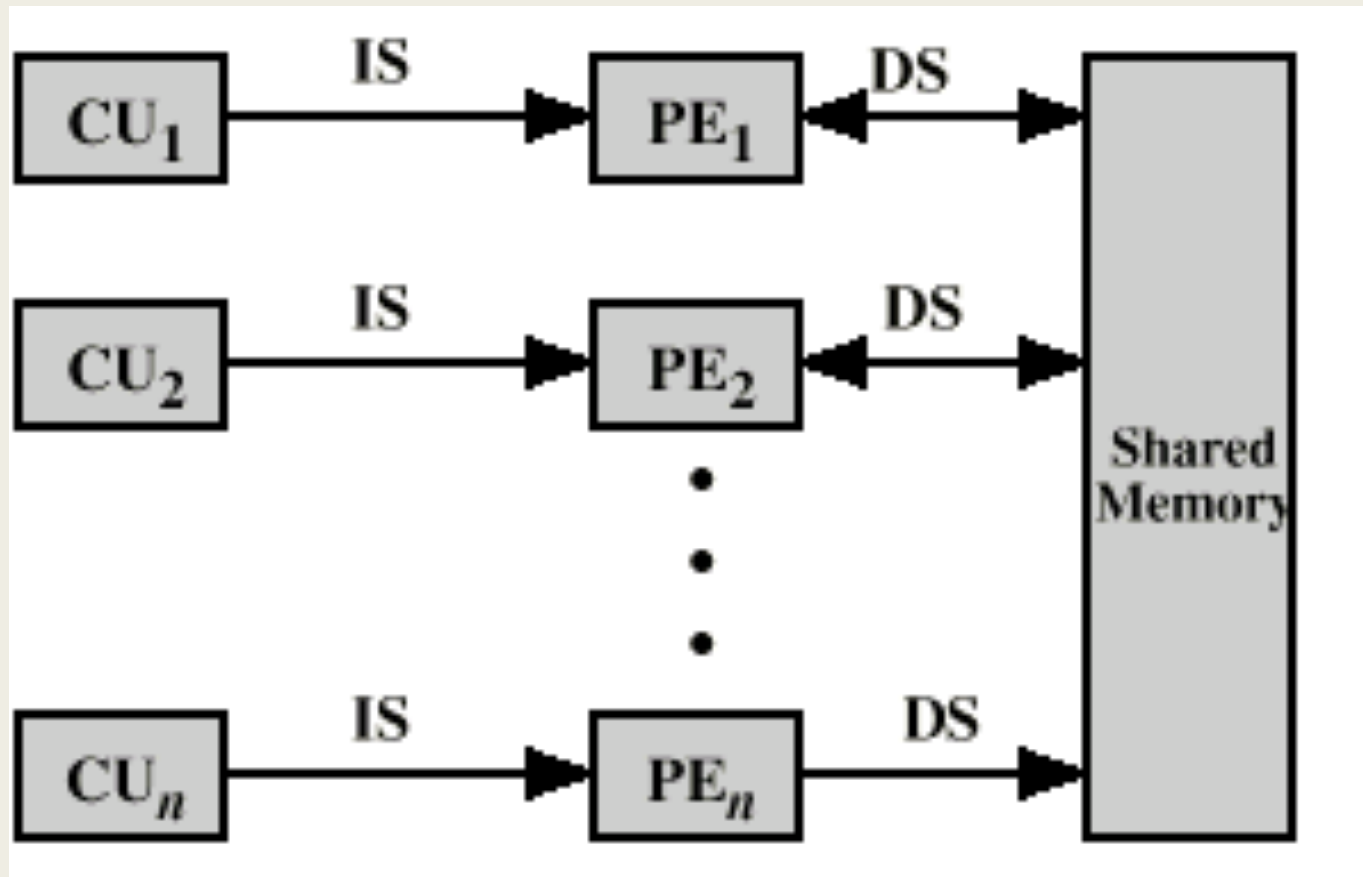
Parallel Organizations - SISD



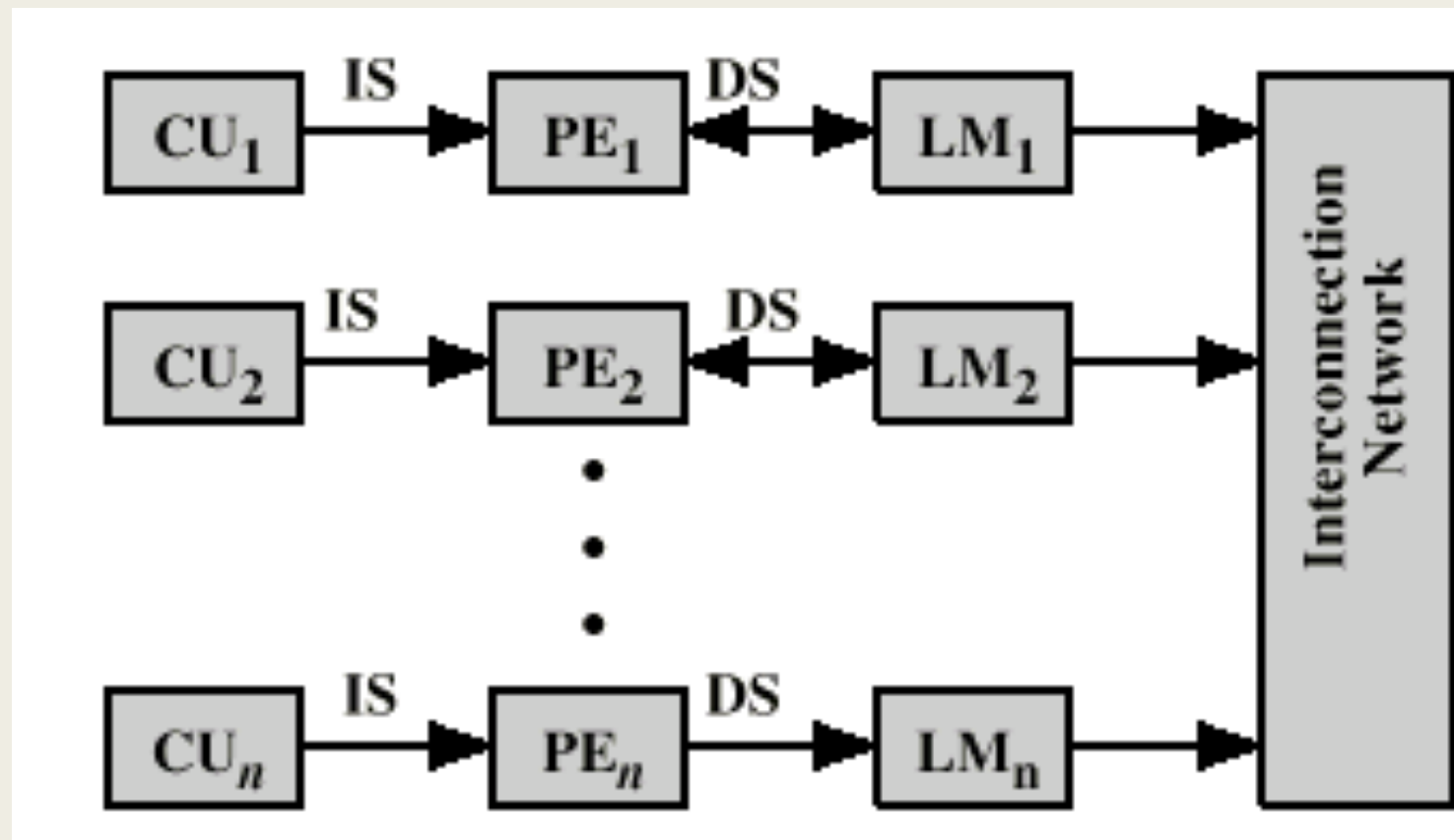
Parallel Organizations - SIMD



Parallel Organizations - MIMD Shared Memory



Parallel Organizations - MIMD Distributed Memory



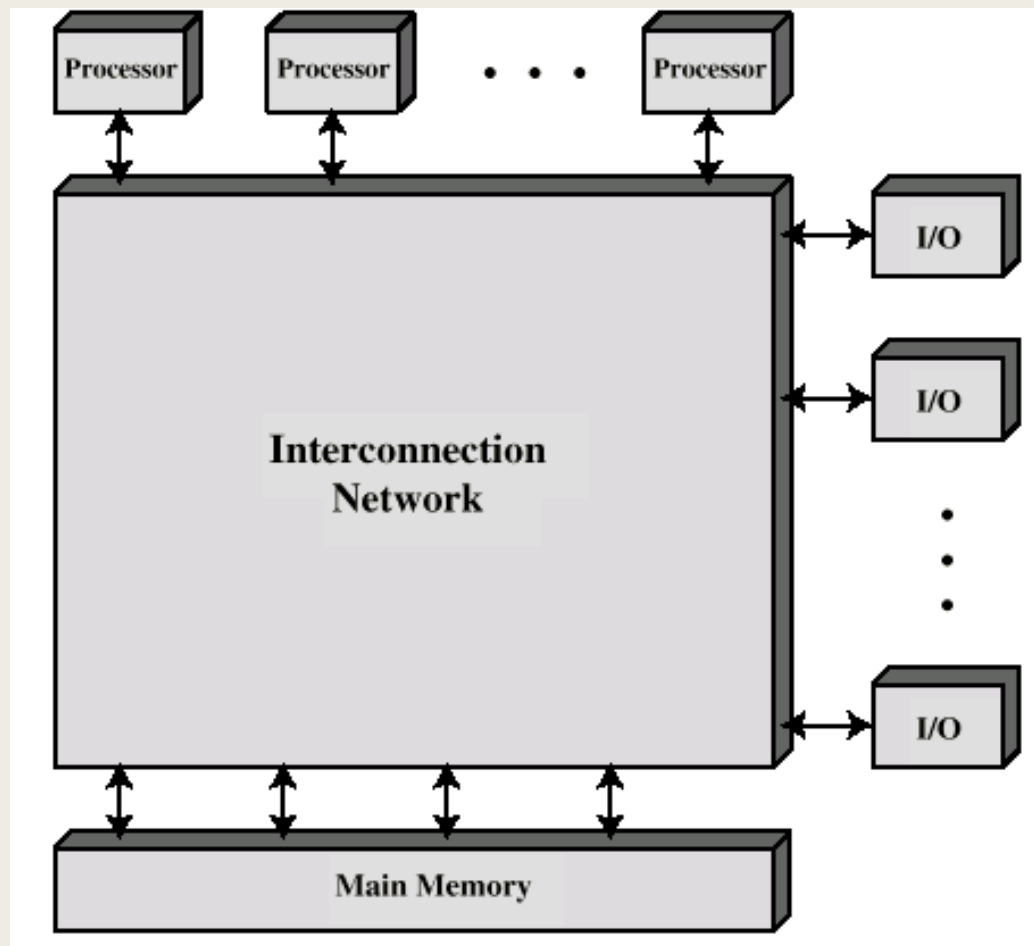
Symmetric Multiprocessors

- A stand alone computer with the following characteristics
 - *Two or more similar processors of comparable capacity*
 - *Processors share same memory and I/O*
 - *Processors are connected by a bus or other internal connection*
 - *Memory access time is approximately the same for each processor*
 - *All processors share access to I/O*
 - Either through same channels or different channels giving paths to same devices
 - *All processors can perform the same functions (hence symmetric)*
 - *System controlled by integrated operating system*
 - providing interaction between processors
 - Interaction at job, task, file and data element levels

SMP Advantages

- Performance
 - *If some work can be done in parallel*
- Availability
 - *Since all processors can perform the same functions, failure of a single processor does not halt the system*
- Incremental growth
 - *User can enhance performance by adding additional processors*
- Scaling
 - *Vendors can offer range of products based on number of processors*

Block Diagram of Tightly Coupled Multiprocessor



Organization Classification

- Time shared or common bus
- Multiport memory
- Central control unit

Time Shared Bus

- Simplest form
- Structure and interface similar to single processor system
- Following features provided
 - *Addressing - distinguish modules on bus*
 - *Arbitration - any module can be temporary master*
 - *Time sharing - if one module has the bus, others must wait and may have to suspend*
- Now have multiple processors as well as multiple I/O modules

Time Share Bus - Advantages

- Simplicity
- Flexibility
- Reliability

Time Share Bus - Disadvantage

- Performance limited by bus cycle time
- Each processor should have local cache
 - *Reduce number of bus accesses*
- Leads to problems with cache coherence
 - *Solved in hardware - see later*

Multiport Memory

- Direct independent access of memory modules by each processor
- Logic required to resolve conflicts
- Little or no modification to processors or modules required

Multiport Memory - Advantages and Disadvantages

- More complex
 - *Extra login in memory system*
- Better performance
 - *Each processor has dedicated path to each module*
- Can configure portions of memory as private to one or more processors
 - *Increased security*
- Write through cache policy

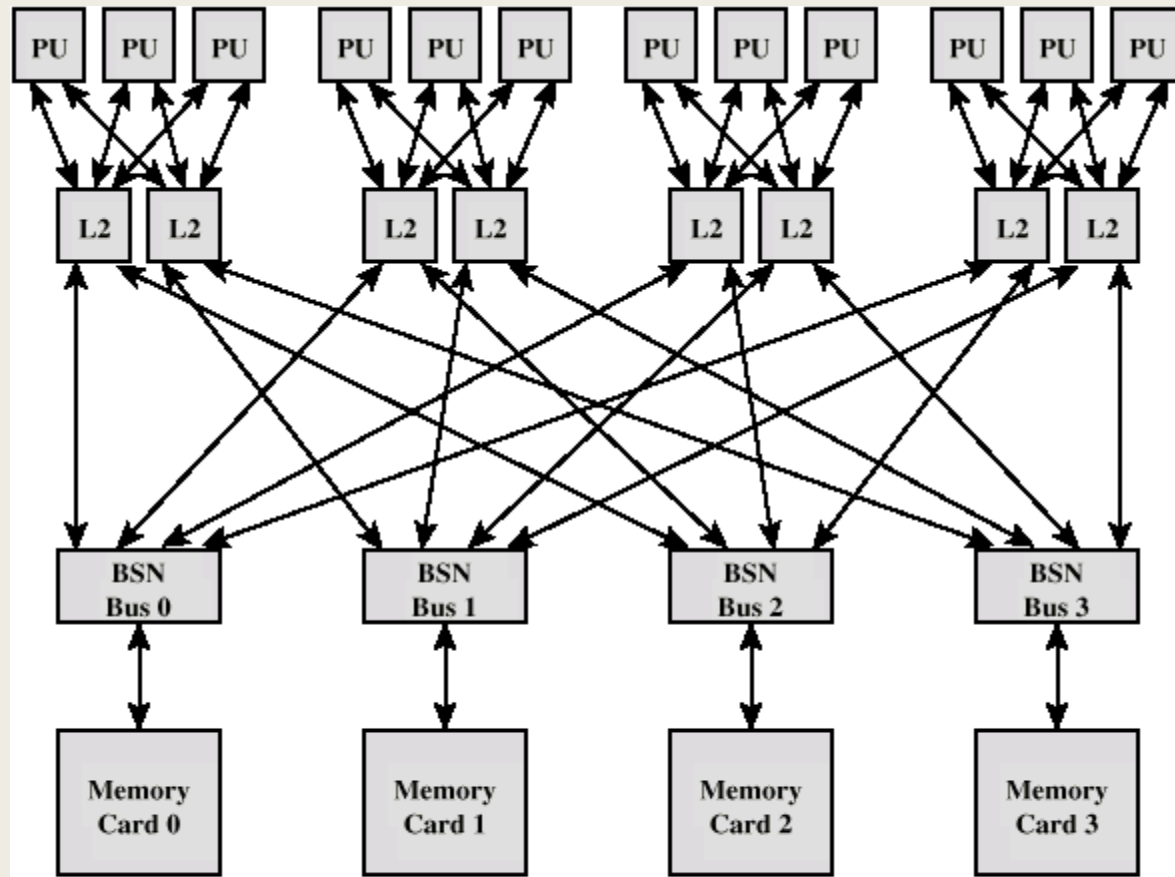
Central Control Unit

- Funnels separate data streams between independent modules
- Can buffer requests
- Performs arbitration and timing
- Pass status and control
- Perform cache update alerting
- Interfaces to modules remain the same
- e.g. IBM S/370

Operating System Issues

- Simultaneous concurrent processes
- Scheduling
- Synchronization
- Memory management
- Reliability and fault tolerance

BM S/390 Mainframe SMP



S/390 - Key components

- Processor unit (PU)
 - *CISC microprocessor*
 - *Frequently used instructions hard wired*
 - *64k L1 unified cache with 1 cycle access time*
- L2 cache
 - *384k*
- Bus switching network adapter (BSN)
 - *Includes 2M of L3 cache*
- Memory card
 - *8G per card*

Cache Coherence and MESI Protocol

- Problem - multiple copies of same data in different caches
- Can result in an inconsistent view of memory
- Write back policy can lead to inconsistency
- Write through can also give problems unless caches monitor memory traffic

Software Solutions

- Compiler and operating system deal with problem
- Overhead transferred to compile time
- Design complexity transferred from hardware to software
- However, software tends to make conservative decisions
 - *Inefficient cache utilization*
- Analyze code to determine safe periods for caching shared variables

Hardware Solution

- Cache coherence protocols
- Dynamic recognition of potential problems
- Run time
- More efficient use of cache
- Transparent to programmer
- Directory protocols
- Snoopy protocols

Directory Protocols

- Collect and maintain information about copies of data in cache
- Directory stored in main memory
- Requests are checked against directory
- Appropriate transfers are performed
- Creates central bottleneck
- Effective in large scale systems with complex interconnection schemes

Snoopy Protocols

- Distribute cache coherence responsibility among cache controllers
- Cache recognizes that a line is shared
- Updates announced to other caches
- Suited to bus based multiprocessor
- Increases bus traffic

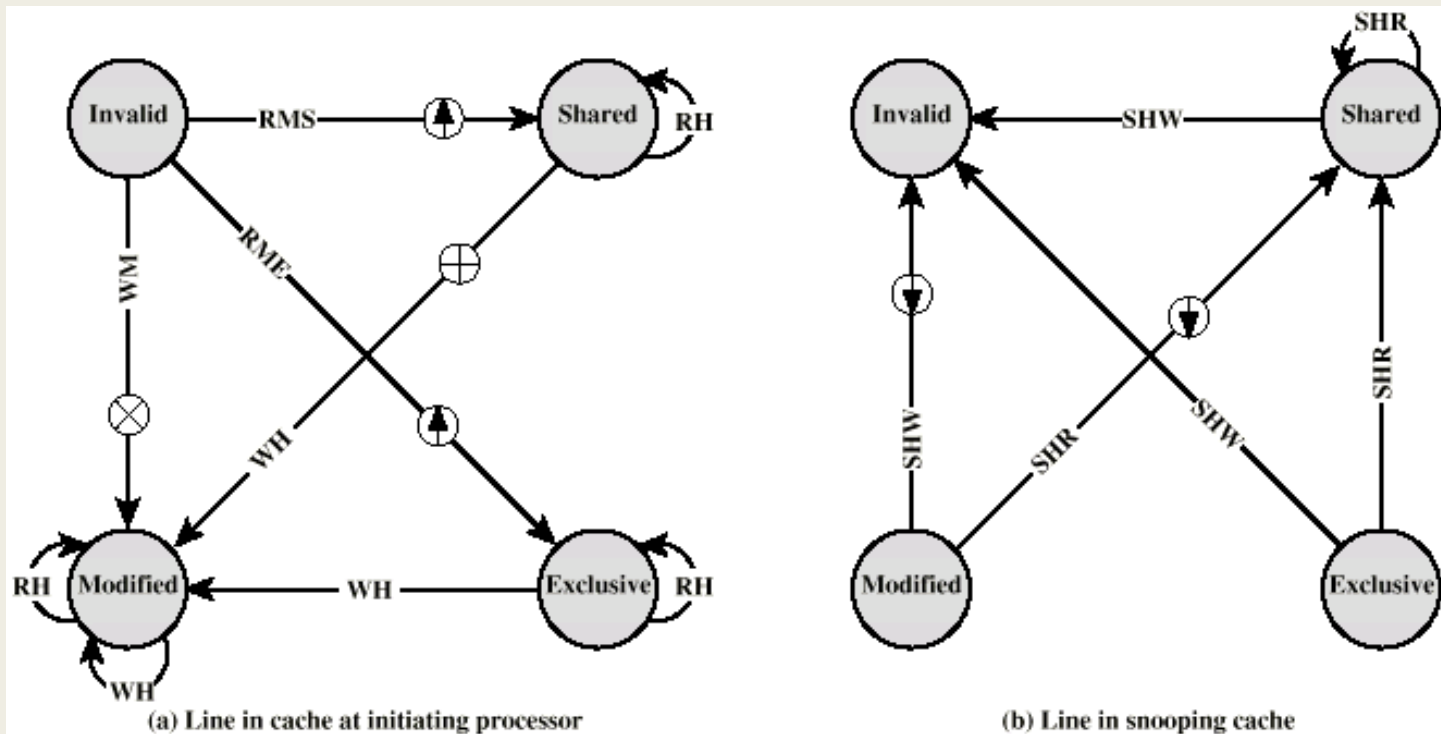
Write Invalidate

- Multiple readers, one writer
- When a write is required, all other caches of the line are invalidated
- Writing processor then has exclusive (cheap) access until line required by another processor
- Used in Pentium II and PowerPC systems
- State of every line is marked as modified, exclusive, shared or invalid
- MESI

Write Update

- Multiple readers and writers
- Updated word is distributed to all other processors
- Some systems use an adaptive mixture of both solutions

MESI State Transition Diagram



RH	Read hit		Dirty line copyback
RMS	Read miss, shared		
RME	Read miss, exclusive		Invalidate transaction
WH	Write hit		
WM	Write miss		Read-with-intent-to-modify
SHR	Snoop hit on read		
SHW	Snoop hit on write or read-with-intent-to-modify		Cache line fill

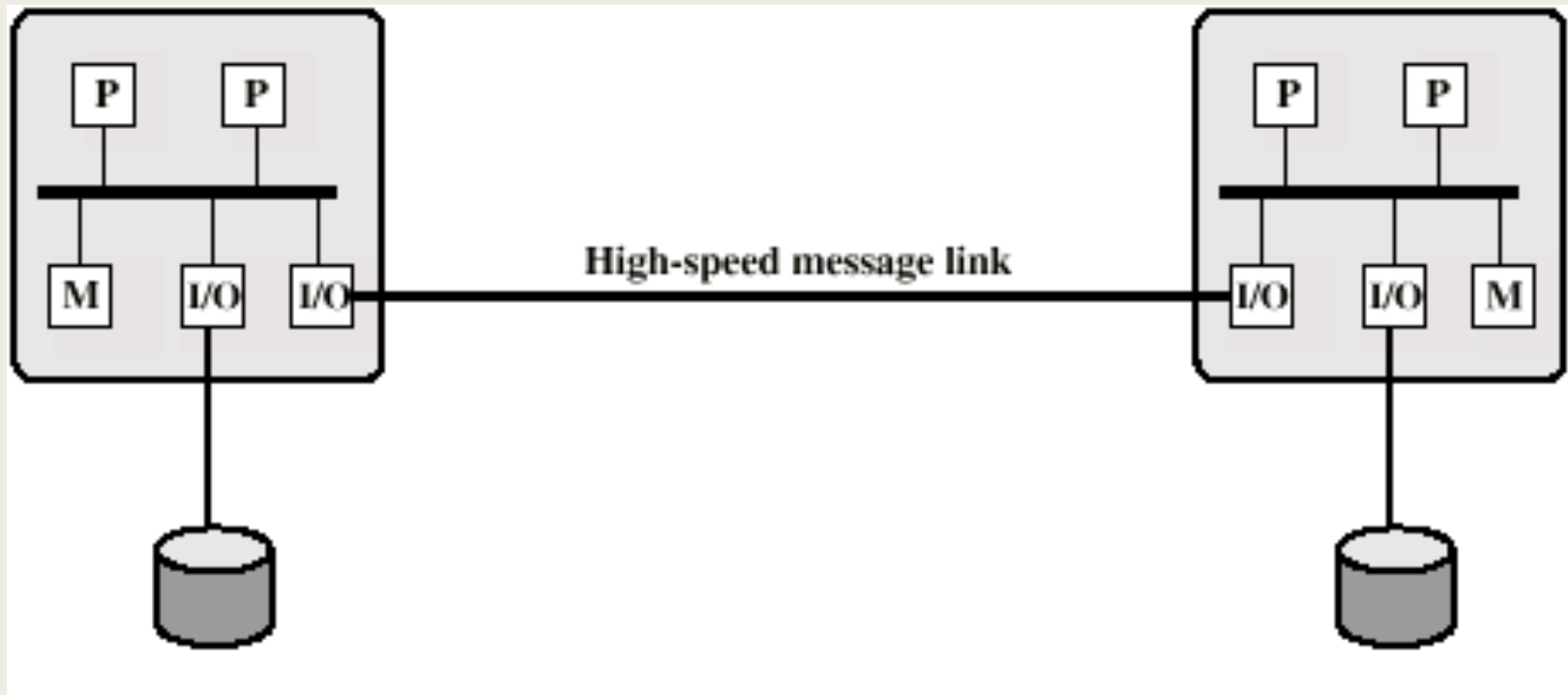
Clusters

- Alternative to SMP
 - High performance
 - High availability
 - Server applications
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- A group of interconnected whole computers
 - Working together as unified resource
 - Illusion of being one machine
 - Each computer called a node

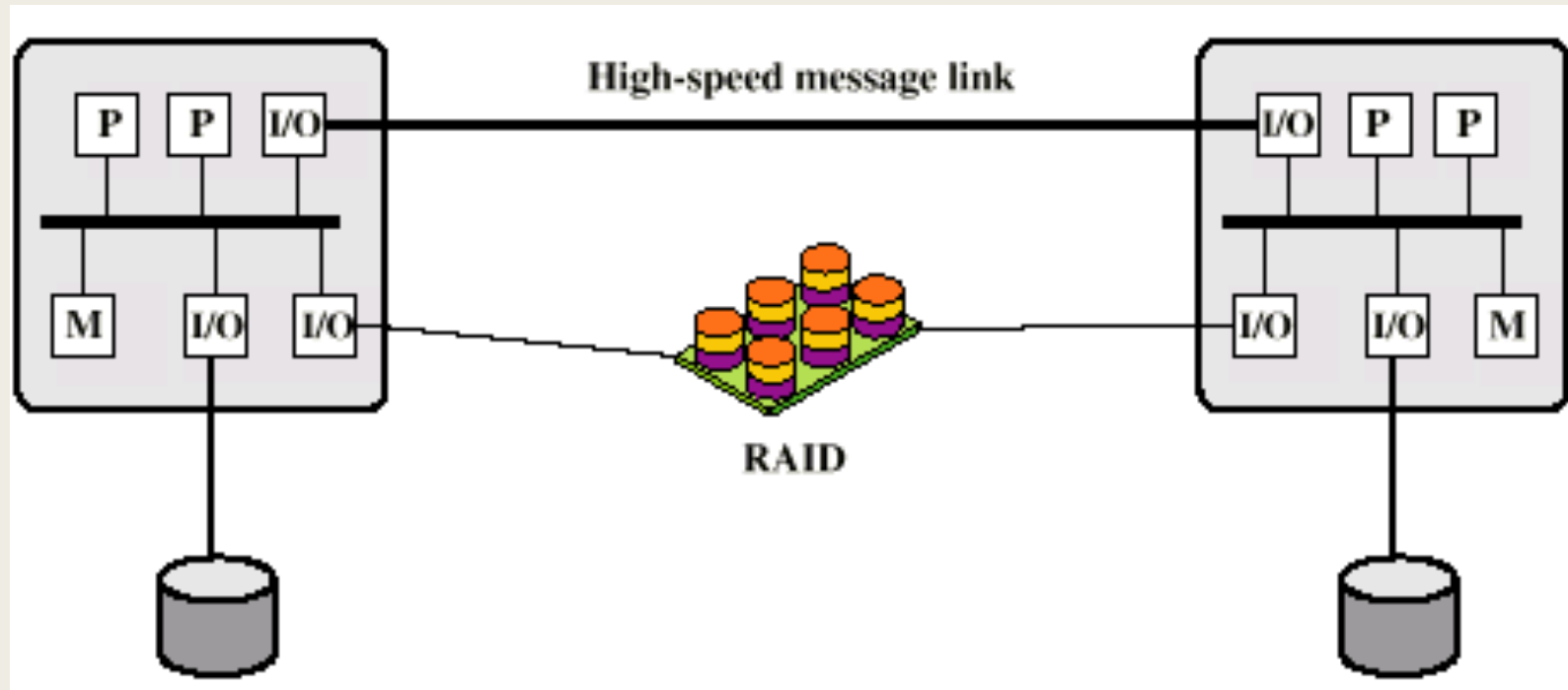
Cluster Benefits

- Absolute scalability
- Incremental scalability
- High availability
- Superior price/performance

Cluster Configurations - Standby Server, No Shared Disk



Cluster Configurations - Shared Disk

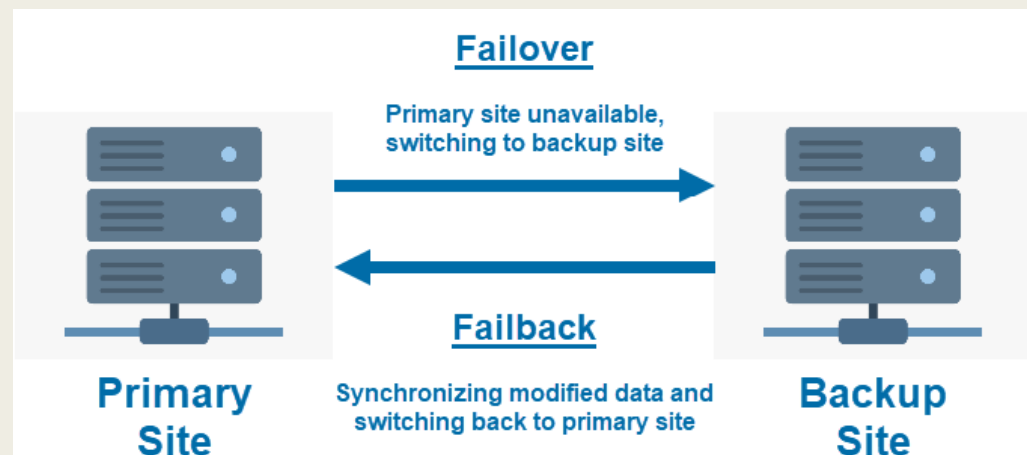


Cluster Configurations

- Passive standby
- Active secondary
- Separate servers
- Servers connected to disks
- Servers share disks

Operating Systems Issues

- Failure management
 - *Highly available*
 - *Failover*
 - *Failback*
- Load balancing



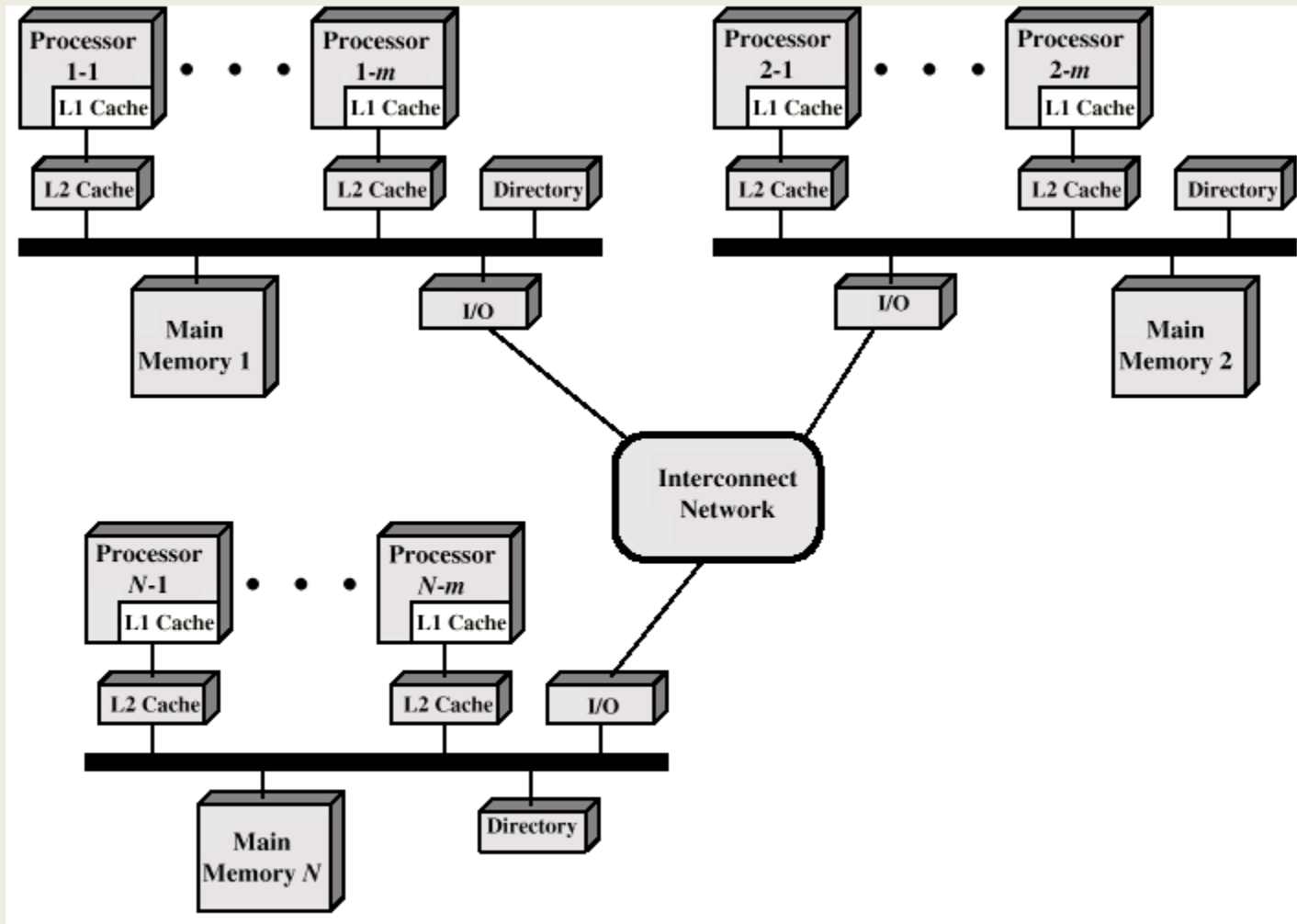
Clusters v SMP

- Both use multiple processors for high demand applications
- SMP is easier to manage
- SMP takes less physical space and less power
- SMP established and stable technology
- Clusters are better for incremental and absolute scalability
- Clusters are better for availability

Non-Uniform Memory Access NUMA

- Uniform memory access
 - *All processors have access to all parts of main memory*
 - *Access time to all regions of memory the same*
 - *Access time by all processors the same*
- Non-uniform memory Access
 - *All processors have access to all memory using load and store*
 - *Access time depends on region of memory being accessed*
 - *Different processors access different regions of memory at different speeds*
- Cache-coherent NUMA
 - *Cache coherence is maintained*

CC-NUMA Organization



NUMA Pros and Cons

- Effective performance at higher level of parallelism than SMP
- Not transparently like SMP
 - *Need software changes*
- Availability

Required Reading

- Stallings Chapter 16